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- The diagram shows a cross-section of a chloroplast. The outer boundary is a double membrane. Inside, the fluid-filled space is labeled 'stroma'. Several stacked, disc-like structures called thylakoids are shown. The space inside a thylakoid is labeled 'thylakoid space'. The number '8' is written near the thylakoids.

PapaCam
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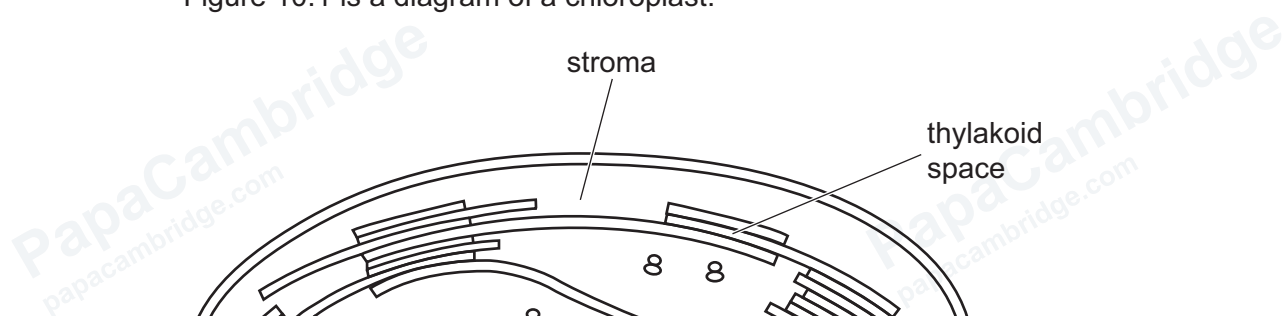


Figure 10.1

With reference to Figure 10.1, outline the relationship between the light-dependent reaction and the Calvin cycle in a chloroplast in photosynthesis.

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(b) Chloroplast pigments can be separated and identified by a technique called chromatography.

A student carried out an investigation using paper chromatography to separate the pigments in a solution containing two chloroplast pigments.

The results of the investigation are shown in Figure 10.2.

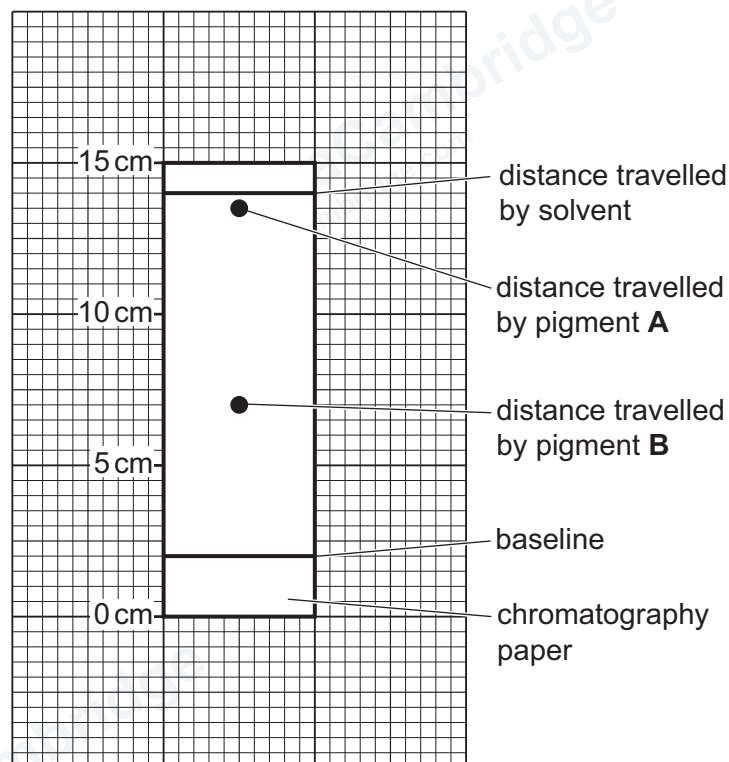


Figure 10.2

(i) The R_f value of pigment **A** is 0.96.

Using Figure 10.2, calculate the R_f value of pigment **B**.

R_f value of pigment **B** [1]



- (ii) Table 10.1 shows the R_f values of some photosynthetic pigments.

Table 10.1

pigment	R_f value
chlorophyll a	0.59–0.68
chlorophyll b	0.40–0.54
carotene	0.94–0.98
xanthophyll	0.15–0.28

Use Table 10.1 to identify pigment **A** and pigment **B**.

pigment **A**

pigment **B**

[1]

- (iii) Describe the role of pigments **A** and **B** in photosynthesis.

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..... [2]

[Total: 8]